

TABLE I
MCU BASED DEVELOPMENT BOARDS

Name	Architecture	Flash memory	Clock speed	Features	Make	Figure
Arduino Due	ARM	512kB	84MHz	AT91SAM3X8E MCU, 54 3.3V I/O, 12 analogue inputs, four UARTs, two DAC outputs, two USBs	Atmel	
STM32F4- Discovery	ARM	1MB	—	MEMS audio sensor, 3-axis digital output accelerometer, USB OTG FS	ST	
LPCXPRESSO Devboard, LPC1549	ARM	256kB	72MHz	Based IDE and GNU C/C++ + tool chain available in free and pro versions, integrated JTAG debugger	NXP	
MSP-EXP430G2 LaunchPad	MSP430	16kB	16MHz	Integrated USB-powered flash emulation tool	TI	
ATSAMC21J18A XPLAINED PRO	ARM	256kB	48MHz	Embedded debugger, digital I/O, CAN/LIN transceiver, Xplained Pro extension	Atmel	
FRDM-KL25Z ARM, KINETIS, KL25Z, FREEDOM	ARM	128kB	48MHz	MKL25Z128VLK4 80 LQFP MCU, Arduino-style headers, cap-touch slider, accelerometer, open SDA	Freescale	

One such path-breaking philosophy surrounds Arduino's business model, which goes like this: "It does not matter anymore whether your product is open source, someone in another country is going to open it up and reverse-engineer it anyway."

So how does one sustain in this volatility? "Basically, what we have is the brand," says Tom Igoe, an associate professor at Interactive Telecommunications Program at New York University, USA.

Looks. Steve Jobs once said, "I do not need engineers. I need artists." What is a product that is low on looks, usability and visual appeal? Sophisticated looks and smart finishing contribute to mass appeal even for a product meant for a technical audience.

Clarity on user segment/intended apps. Very often than not, it is noticed that people stuff development boards with anything and everything since it is not too clear what the hardware is meant to do and what sort of audience does it cater to. This is a direct contributor to the fact that most companies are unable to come up with cost-effective hardware. Moreover, we are often unaware of the real computational capabilities of the board.

Avoiding premature optimisation. Optimisation is something

that you should keep back for the end product. It is very common for the project scope to be refined or redefined at every stage of your project. Due to client demand for more flexibility in the product we have to keep a lot of related options open while freezing the development board instead of trying to optimise prematurely.

MCU based development boards

Some development boards are general microcontroller (MCU) specific and are available from semiconductor manufacturers. These boards include Arduino boards, STM32-Discovery kits, TI Launchpad (MSP-EXP430G2), Xplained boards from Atmel and others. Additionally, some economically-available boards are designed by local companies.

In both cases, MCU development boards come in two flavours, one being MCU evaluation boards where focus is on evaluating MCU capabilities. Second category is application/feature specific where the board is designed for the purpose of evaluating certain features of the MCU and is usually focussed on some specific application area.

Single-board computers

Today, single-board computers

(SBCs) can be grouped into two main categories: proprietary and open source.

Proprietary SBCs are generally designed for use in end applications or as a reference to be evaluated. These are often industrialised designs that have gone through the same type of testing that an end product requires and are often integrated into end product designs.

Open source SBCs, on the other hand, offer users access to both the hardware design and layout as well as access to the source code used on the board. This is ideal for all users as they can easily understand how the software and hardware operates and adopt the design to meet their end designs requirements or simply learn how a piece of hardware or software works.

Current SBCs come with a wide variety of processor types, most with GPUs onboard. These processors range from X86 based processors from the traditional PC space (AMD and Intel) to ARM processors that have traditionally been used in the industry and more recently mobile spaces.

The most prevalent form of software used on SBCs is Linux with numerous derivations including Android, Ubuntu, Fedora, Debian and Arch Linux, as well as FreeBSD and